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PEE 515 — Past Questions & Materials
by Renance

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Paper 1 of 1 — PEE 515 - Natural Gas Engineering II

Level 500 • First Semester • Handout

PNG 516

Natural Gas Engineering II

Ogbon Natural Gas Engineering II Slide #1

Pipeline Design

□ Flowline Design

□ Operating Pressure P(psia)

□ This is the amount of pressure required to push, force or flow the gas through the entire pipeline length to the destination.

□ It is meant to overcome the weight of the fluid and all frictional losses

□ Maximum Allowable Operating Pressure (MAOP)

□ This is the maximum allowable operating pressure for the pipeline

□ It is based on Material of Construction (MOC) and thickness of the vessel/pipeline

The MAWP will be changing with time because of corrosion and vessel fatigue.

□ Design Pressure

□ This is the design operating or rating pressure for the pipeline.

□ This is the pressure required to sustain the stress imposed on it from the external and the internal pressures during operation.

□ Design pressure depends on the kind of system (fuel, water, steam, etc.) that is used in the vessels.

□ Design pressure is always equal or less than the MAWP of the vessel or equipment. $PD \leq MAWP$

□ The vessel or pipeline relief pressure setting is based on design pressure, and it can be equal to the MAWP but not higher.

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Pipeline Design

□ Flowline Design

□ Reynolds Number

□ Designated as N_{re} or Re

□ General for Liquids - field units

Where

$N_{Re} = 928 \, dv \, d$ - inches

v - ft/sec

μ - cp

ρ - lb/gal

QI = bbl/day

SG = specific gravity relative to water

□ For Gases - field units

Where;

μ = viscosity, cp,

d = pipe inside diameter, in.,

S = SG of gas at standard conditions relative to air

Q_g = gas-flow rate, MMscf/D.

□ Flow Type

□ Turbulent, $N_{Re} \geq 2000$ result in greater pressure loss

□ Laminar, $N_{Re} \leq 2000$

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Pipeline Design

□ Flow Line Design

The basic and effective equations commonly employed to calculate the gas flow in pipes in the oil and gas industry include;

□ Darcy-Weisbach Equation

□ Weymouth Equation

□ Panhandle Equation and

□ Spitzglass Equation

□ Darcy-Weisbach Equation – Incompressible Model

□ The Darcy-Weisbach equation is basically for incompressible fluid but can be applied in gas flow calculations under certain conditions.

□ Crane (1988) states that if the pressure drop is less than 10% of P_1 you can use the incompressible model, BUT the gas density should be based on either the upstream or the downstream conditions.

□ For a short pipe or low flow with pressure drop between 10% and 40%, of P_1 the Darcy-Weisbach may be more accurate and it can then be applied by using density based on the average of the upstream and downstream conditions.

□ If the pressure drop exceeds 40%, then use a compressible model, like the Weymouth, Panhandle, etc.

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Pipeline Design

□ Flow Line Design

□ The Compressible Models

□ Each of the compressible models equation has its own special representation of the friction factor to allow the equations to be solved analytically irrespective of the fact that they were developed from the fundamental energy equation for compressible flow.

□ The accuracy and applicability of each equation fall within certain ranges of flow and pipe diameter and these will be highlighted in the treatment of

each equation below

□ Apart from the Darcy-Weisbach equation, the other equations were developed for turbulent flow in long pipelines.

□ For low flows, low pressures, or short pipes, they may not be applicable.

□ If the pressure drop in a pipeline is less than 40% of P_1 , the Darcy-Weisbach may be more accurate than the Weymouth or Panhandles for a short pipe or low flow

□ If the pressure drop exceeds 40% of P_1 , then use a compressible model, like the Weymouth, Panhandle, etc. that will be more accurate

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Pipeline Design

□ Flow Line Design

□ Darcy-Weisbach Equation for Frictional Flow Pressure Loss

L - Feet

→ Field Units with Flow Rate Q d - inches

□ L - Fluid specific gravity

□ $P \propto 0.0000115 f$

□ $q \propto L \propto g$

2

L qL - flow rate in bbl/day

f

$d^5 f$ - Moody friction factor (Darcy friction factor)

→ Field Units with Velocity V

f = Moody friction factor, dimensionless,

L = pipe length, ft,

D = pipe diameter, ft,

V = velocity, ft/sec,

g = gravitational constant ft/sec²,

ΔP = pressure drop, psi,

ρ = density, lbm/ft³, and

→ SI Units

f = Moody friction factor,

2

$L \propto u$ dimensionless,

□ $Pf \propto f L$ = pipe length, m,

$2d D$ = pipe diameter, m,

V = velocity, m/sec,

g = gravitational constant m/sec²,

ΔP = pressure drop, Pas,

ρ = density, kg/m³, and

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Pipeline Design

□ Flow Line Design

□ General Darcy-Weisbach Equation for both Liquid and Gas

□ Units Consistent for both SI and imperial units

f = Moody friction factor - dimensionless

L = pipe length - ft or m

D = Pipe diameter - ft or m

v = fluid velocity - ft/sec or m/sec

h_f = head loss -ft or m of water

g = acceleration due to gravity = 32.174

ft/s² = 9.806 m/s².

Q = ft³/sec or m³/sec

□ = density - lb/ft³ or kg/m³

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Pipeline Design Class

Exercise

□ Flow Line Design

□ General Darcy-Weisbach Equation for both Liquid and Gas

Calculate the gas flow using Darcy-Weisbach model for the pipeline flow with the following; pipe length = 450ft, pipe diameter = 4¾" frictional factor = 0.000612, and ΔP = 15.56ft of water

ΔP □ 0.0000115 f

Δq □ L □

g

2

L

f

d^5

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Pipeline Design

□ Flow Line Design

□ Darcy-Weisbach Equation for Frictional Flow Pressure Loss

For Laminar flow

For $N_{re} < 2,000$, pipe roughness is not a factor in friction determination

Moody friction factor

$f = 64$ and

N_{re}

For Turbulent flow

When $N_{re} \geq 2000$, the flow in the pipe becomes practically turbulent and the value of friction factor (f) then becomes less dependent on the Reynolds but more on the relative roughness (e/D) of the pipe. For rough pipe in turbulent flow you must use the Moody Chart to obtain f or with the use of Colebrook-White equation for gas flow

where, in USCS units,

f – Friction factor, dimensionless

D – Pipe inside diameter, in.

e – Absolute pipe roughness, in.

R – Reynolds number of flow,
dimensionless

For hydraulically smooth pipes such as glass ($e = 0$); for copper and plastic tubing in turbulent flow, use Blasius equation for f ($4,000 < Re < 100,000$)

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Pipeline Design

□ Flow in Gas Pipeline

□ Weymouth equation

□ The Weymouth equation is the most common of the equations probably because it has been around for quite sometime

□ It is mainly for transmission lines and not for gathering and distribution pipelines.

□ The Weymouth equation is recommended for

□ Single phase fluids

□ Diameter pipe ($ID \geq 15''$).

□ Long lengths of segments < 32 km (20 miles)

□ Medium to high pressure (100 psia to 1000

□ Not applicable to low flows, low pressures, or short pipes

□ Pipes within production batteries

□ Branch gathering lines,

□ Medium- to high-pressure (± 100 psig to $> 1,000$ psig) applications,

□ High Reynolds number flow where the Moody friction factor is merely a function of relative roughness.

□ The Weymouth equation is ideal for designing branch laterals and trunk lines in field gas-gathering systems.

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Pipeline Design

□ Flow in Gas Pipeline

□ Weymouth equation

□ The Weymouth equation is given below in both lengths of miles and feet

□ In Miles L - length in Miles

□ d - pipe inside diameter in inches

□ q_g - flow rate in scf/day

Z - Average compressibility factor

P_1 - upstream pressure in psia

P_2 - downstream pressure in psia

T - Average gas inlet temperature in °R

E - pipeline efficiency

Δz - Elevation of the pipe

between exit and

entrance in feet

When $\Delta Z = 0$, $S = 0$, $L_e = L$

and $e_s = 1$

and equation becomes

$T_b 2.667 \left[\frac{P_1}{P_2} \right]^2$

2

$q_g = 433.49 d^{2.667} \left[\frac{P_1}{P_2} \right]^2$

$P_b = L \left[\frac{Z T}{E} \right]$

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Pipeline Design

□ Flow in Gas Pipeline

□ Weymouth equation

□ The Weymouth equation is given below in both lengths of miles and feet

L - length in feet

d - pipe inside diameter in inches

S - Fluid specific gravity

□ In Feet q_g - flow rate in mm scf/day

Z - Average compressibility factor

P_1 - upstream pressure in psia

P_2 - downstream pressure in psia

T_1 - Average gas inlet temperature in °R

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Pipeline Design

□ Flow in Gas Pipeline

□ Weymouth equation

□ Multiphase flow using Weymouth equation is similar but pretty lengthy and methods include

- ☐ Computer solutions to multiphase flow
- ☐ Analogies with existing line
- ☐ Horizontal flow Gradient curves
- ☐ Calculate the flow capacity using Weymouth

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Pipeline Design

- ☐ Flow in Gas Pipeline
- ☐ The Panhandle Equation
- ☐ The Panhandle equation is used for moderate-Reynolds-number flows where the Moody friction factor is independent of relative roughness and is a function of Reynolds number.

- ☐ It incorporates the gas properties associated with natural gas into the general energy balance

• E = efficiency factor equation

– (new pipe: 1.0; good operating conditions: 0.95; average operating conditions: 0.85),

L_m - length in miles

d - pipe inside diameter in inches

S - Fluid specific gravity

q_g - flow rate in mm scf/day

Z - Average compressibility factor

P₁ - upstream pressure in psia

P₂ - downstream pressure in psia

- ☐ Long lengths of segments T₁ - Average gas inlet temperature in OR
- ☐ Not applicable to low flows, low pressures, or short pipes
- ☐ It has a higher degree of accuracy for higher flowrates and smaller pipe diameter

- ☐ It also produces a more accurate answer when incorporating an elevation change

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Pipeline Design

- ☐ Flow in Gas Pipeline
- ☐ The Panhandle Equation
- ☐ Other versions of the equations with length in feet include
- ☐ The Modified Panhandle Equation A
- ☐ This equation also incorporates the gas properties associated with natural gas into the general energy balance equation. It has an efficiency factor for the Reynolds number within a range of 5 to 11 million. While pipe roughness is not in the equation, the efficiency

factor in the equation takes into account inefficiencies caused by the roughness of the pipe

When $\epsilon Z = 0$, $S =$

0, $L_e = L$ and e_s

$= 1$ and equation

becomes

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Pipeline Design

□ Flow in Gas Pipeline

□ The Panhandle Equation

□ Other versions of the equations with length in feet include

□ The Modified Panhandle Equation B

□ This is a slight modification of the original equation which allows the equation to be accurate for a larger Reynolds number range of about 4 to 40 million

When $\epsilon Z = 0$, $S =$

0, $L_e = L$ and e_s

$= 1$ and equation

becomes

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Pipeline Design

□ Flow in Gas Pipeline

□ Spitzglass Equation

□ This equation is recommended for low-pressure vent lines < 12 in. in diameter ($\Delta P < 10\%$ of P_1).

L - length in feet

d - pipe inside diameter in inches

S - Fluid specific gravity

q_g - flow rate in mm scf/day

Z - Average compressibility factor

□ h_w - Pressure head in inches

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Pipeline Design

□ Flow in Gas Pipeline

Example

Calculate the capacity using both the Weymouth and modified Panhandle equations for the following flow conditions:

$P_1 = 600$ psia, $P_2 = 100$ psia, $T = 90$ OF, $\gamma = 0.67$, $Z = 0.89$, $d = 2$ inches and $L = 52,800$ ft

□ Using the Weymouth Equation

0.5 0 .5

$T \square P^2 \square P_{22} \square 520 \cdot 2.667 \square \square 600^2 \square 100^2 \square$

$qg \square 433.49 \cdot b \cdot d \cdot 2.667 \square 1 \square \square 433.5^2 \square$

$P_b \square L \square Z \cdot T \square 14.7 \square 10,000 \cdot x^{0.67} \cdot x^{0.89} \cdot 550 \square$

$\square \square \square$

$\square 31,815.93 \text{ scf / day}$

□ Using the Panhandle Equation

1.0788 0.5392 1.0788 0.5392

$\square T \square \square P^2 \square P^2 \square \square 520 \square 2.6182 \square \square 600^2 \square 100^2 \square$

$q \cdot g \square 435.7 \square \square b \square \square d \cdot 2.6182 \square 0.18539^2 \square \square 435.7 \square \square 2 \square$

$\square \square TLZ \square \square \square 0.67 \cdot 0.8539 \cdot 550 \cdot 10,000 \cdot x^{0.89} \square$

$\square P_b \square \square \square 14.7 \square \square \square$

$\square 36.340.414 \text{ scf / day}$

□ Using the Modified Panhandle Equation B

1.02 0.51 1.02 0.51

$\square T \square \square P^2 \square P^2 \square \square 520 \square 2.53 \square \square 600^2 \square 100^2 \square$

$q \cdot g \square 737 \square \square b \square \square d \cdot 2.53 \square 0.1961^2 \square \square 737 \square \square 2 \square$

$\square \square TLZ \square \square \square 0.67 \cdot 0.961 \cdot 550 \cdot 10,000 \cdot x^{0.89} \square$

$\square P_b \square \square \square 14.7 \square \square \square$

$\square 52,416.224 \text{ scf / day}$

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Pipeline Design

□ Design Factors

□ General Design Factor(F) - 0.6 or 0.8

Pipeline design Location Factor(L) Pipeline Design Welding Factor(J)

Pipeline Specification/Grade:

Temperature Derating Factor(T) API %L Grades

for Steel Pipe

No. Temperature Temp. Derating

0

F Factor

1. 250 or less 1.000

2. 300 0.967

3. 350 0.933

4. 400 0.900

5. 450 0.867

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Pipeline Design

□ Pipe Wall Thickness

DP where,

t = Pipe wall thickness, mm

P = design pressure, psig

D = inside diameter of pipe in mm

Y = minimum yield strength, psig

F = design factor

L = Location factor based on class location

J = welding joint factor

T = Temp. derating factor

where

□ Factors to be Considered in Pipeline Implementation

1. Market survey - Present & future demand
2. MAOP & Design pressure
3. Map of the preliminary selected pipeline route (L)
4. Pipe Size (D)
5. Wall thickness (t)

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Pipeline Design and Construction

□ Factors to be Considered in Pipeline Implementation

6. Pipe specification / grade
7. Bill of materials
8. Total cost estimate
9. Selection of ROW
10. Detailed survey and Preparation of alignment drawings

□ Load consideration:

□ Consider the present load as well as the assumed gas load of the that particular area for the next 20 years (effective life of the pipeline)

• Ono 21

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Pipeline Design and Construction

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Pipeline Design and Construction

• Ono 23

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Transmission Lines Modification

□ During the life time of pipeline network there comes a time when the system becomes constrained for throughput or inventory capacity.

□ Older pipelines which were designed with additional reserve capacity,

hardly requires upgrading.

- But pipeline systems designed for a precise load profile, soon or later needs upgrades to accommodate the new flow demands.

- This has resulted in re-designing of newly commissioned pipeline system at the last minute, or being looped they are a few years old.

- before

- The design of modern commercial pipeline systems and an increase in their capacities, is tailored precisely to the known customers demand in a bid to avoid added capital expense.

- Therefore the advent of pipeline system increase capacity project is Invariably a commonplace in the natural gas industry.

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Transmission Lines Modification

- Increasing pipeline capacity

Methods employed for increasing pipeline system throughput include upgrade projects in pipeline looping, compression and maximum allowable operating pressure (MAOP).

- Additional compression capacity

- Increase in throughput by addition of more compression (additional compressors) is the most common form of capacity increase

- It is normally accomplished with minimal interruption to existing operations – about 18 months.

- Throughput increases of 5–20 per cent can be achieved without additional pipelines

- The only limitations are space for compressors installation and dependent on the mechanical reliability of the units and pipelines.

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Transmission Lines Modification

- Increasing pipeline capacity

- Increasing MAOP

- The current revision of AS2885.1 now gives the possibility to increase the MAOP of a pipeline which at present limits the increase up to a design factor of 0.72,

- This allows some pipelines which have been operating at lower pressures than their original design pressure to increase throughput and MAOP of such line can potentially be increased provided the pipeline has been constructed correctly, hydrotested to the correct pressure and maintained well.

- The process involves an engineering exercise to demonstrate the new

MAOP is safe, and these may include

- Intelligent pigging process.
- Modifications to pipeline to replace lower pressure capacity components or questionable components.
- A MAOP upgrade can be achieved in 9–18 months and can achieve increments up to around 10 per cent of throughput.

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Looped Pipeline System

- Pipeline Looping

This is the installation of a parallel pipeline adjacent to the existing line to lower the flow resistance and hence the pressure drop, and to thus increase the throughput.

- Considering the complex issues of compatibility and cost associated with compression, the major alternative to additional compression is pipeline looping.
- In looping, the length and diameter, and placement of each loop section can be precisely tailored to suit the new load profile.
- Increases in throughput of over 200 per cent are readily accomplished if the diameter and length of the loops are sufficient BUT Looping has been considered to be more expensive than compression addition for flow increases which are less than 50 per cent of free flow
- Pipelines can also be looped in stages to suit increasing demands on a defined timetable.
- The duration for looping is dependent on the amount of pipe to be installed, so can vary from about 12 to 24 months or longer.

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Looped Pipeline System

- Campbell's equation
- The figure below is a simple diagram for pipeline looping

K

- The parallel pipe B is looped at point K to divide the old two lengths of A and C
- In the looped pipeline system in the above figure, the length of the parallel pipeline (B) is determined in a way that, in spite of the increase in gas flow rate, the amount of pressure drop in the main pipeline (A+C) was not changed.
- Note that the length and diameter of the parallel pipeline are interrelated as such, only the length or the diameter of the pipeline can be considered as a design parameter.

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Looped Pipeline System

□ Campbell's equation

□ Campbell's basic equation to determine the length of the parallel line to the old line is as follows

2

□ Q □

1 □ □ old □

X_f □ □ Q_{new} □

1 X_f is defined as

1 □ 2

□ 8

□ 1 □ □ DB □ □

3

□ □ D □ □

□ □ □ A □ □

Where X_f □ a fraction of the length of the main pipeline (A+C) which is looped with a parallel pipe B

Q_{old} □ old gas flow rate and

Q_{new} □ new (increased) gas flow rate.

K

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Looped Pipeline System

Campbell's equation

□ Basic Assumptions of Campbell's Equation

□ Horizontal pipeline,

□ Isothermal and steady state flow, and

□ Constant gas compressibility factor

Exercise

A horizontal pipeline with a length of 107.4 kilometers, diameter of 54.76 inches, inlet pressure of 1000 psia, inlet temperature of 45 °C and flow rate of 80 million standard cubic meters per day (MMSCMD).

What length of 42 inch diameter parallel pipeline will be required to increase the gas flow rate up to 110 MMSCMD?

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